

# Benchmarking of NQCH's quantum computer

April 29, 2026

## 1. Report of Changes

**Platform:** sinq20

**Calibration-id:** *ec305fcaf377a05f4fb391a8ee18282bfae18df6*

**Calibration date:** 2026-04-28 07:58:51

**Calibration note:** chore(sinq20): update readout parameters

**Experiment-id:** 20260428212203

**Experiment date:** 2026-04-28 21:22:04

**Experiment note:** *temporary note!!!*

**Platform:** sinq20

**Calibration-id:** *4dc4082f38a53222b3956c22202d32a520d4bc78*

**Calibration date:** 2026-01-15 02:03:03

**Calibration note:** chore(sinq20): 2q gates cal 0-1, 0-3, 2-3, 3-4, 1-4, 4-5, 4-9, 3-8, 8-9

**Experiment-id:** 20260118161538

**Experiment date:** 2026-01-18 16:15:38

**Experiment note:** *temporary note!!!*

## 2. Version Comparison

| Library      | Version | Library | Version |
|--------------|---------|---------|---------|
| qibo         | 0.2.23  | numpy   | 2.4.4   |
| qibolab      | 0.2.9   | qibocal | 0.2.5   |
| matplotlib   | 3.10.8  | scipy   | 1.17.1  |
| scikit-learn | 1.8.0   | pandas  | 2.3.3   |
| networkx     | 3.6.1   | sympy   | 1.14.0  |
| torch        | 2.11.0  |         |         |

| Library      | Version | Library | Version |
|--------------|---------|---------|---------|
| qibo         | 0.2.22  | numpy   | 2.2.6   |
| qibolab      | 0.2.9   | qibocal | 0.2.4   |
| matplotlib   | 3.10.3  | scipy   | 1.15.3  |
| scikit-learn | 1.6.1   | pandas  | 2.2.3   |
| networkx     | 3.4.2   | sympy   | 1.14.0  |
| torch        | 2.7.0   |         |         |

## 3. One and two qubit fidelities

The single qubit fidelity is obtained via Randomized-Benchmarking. The two-qubit fidelity is the "Bell-state fidelity".



#### 4. Statistics

|                | Average | Median | Min   | Max   |
|----------------|---------|--------|-------|-------|
| T1 (ns)        | 23.1    | 24.5   | 13    | 30.5  |
| T2 (ns)        | 11.4    | 11.7   | 2.18  | 23.9  |
| Fidelity       | 0.951   | 0.993  | 0.855 | 0.999 |
| RO Fidelity    | 0.986   | 0.986  | 0.976 | 0.99  |
| 2q RB Fidelity | -       | -      | -     | -     |
| 2q CZ Fidelity | -       | -      | -     | -     |

|                | Average  | Median   | Min   | Max      |
|----------------|----------|----------|-------|----------|
| T1 (ns)        | 1.28e+04 | 1.23e+04 | 646   | 3.65e+04 |
| T2 (ns)        | 2.36e+25 | 3.68e+03 | 125   | 9.43e+26 |
| Fidelity       | 0.99     | 0.997    | 0.887 | 0.999    |
| RO Fidelity    | 0.794    | 0.777    | 0.777 | 0.927    |
| 2q RB Fidelity | 0.965    | 0.972    | 0.923 | 0.981    |
| 2q CZ Fidelity | 0.975    | 0.974    | 0.967 | 0.991    |

#### 5. Best Qubits Selection

| k-qubits | Best Qubits      | Fidelity |
|----------|------------------|----------|
| 2        | 5, 6             | 0.935    |
| 3        | 5, 6, 10         | 0.929    |
| 4        | 5, 6, 10, 15     | 0.929    |
| 5        | 5, 6, 10, 15, 16 | 0.928    |

| k-qubits | Best Qubits   | Fidelity |
|----------|---------------|----------|
| 2        | 2, 3          | 0.981    |
| 3        | 0, 2, 3       | 0.976    |
| 4        | 0, 1, 2, 3    | 0.970    |
| 5        | 0, 1, 2, 3, 4 | 0.965    |

#### 6. Summary metrics

| Benchmark       | Metric   | Value | Runtime    |
|-----------------|----------|-------|------------|
| Mermin          | Max      | 3.358 | 227.53 sec |
| Grover 2q       | Max prob | 0.874 | 3.13 sec   |
| Grover 3q       | Max prob | 0.225 | 2.44 sec   |
| GHZ state prep. | Success  | 0.903 | 3.17 sec   |

| Benchmark       | Metric   | Value  | Runtime  |
|-----------------|----------|--------|----------|
| Mermin          | Max      | -3.316 | 1.02 sec |
| Grover 2q       | Max prob | 0.89   | 2.36 sec |
| Grover 3q       | Max prob | 0.385  | 2.43 sec |
| GHZ state prep. | Success  | 0.910  | 3.95 sec |

#### 7. Randomized Benchmarking Results

| Qubit n | Fidelity | Error Bars | Qubit n | Fidelity | Error Bars |
|---------|----------|------------|---------|----------|------------|
| 0       | 0.995    | ± 0.000224 | 10      | 0.996    | ± 0.00012  |
| 1       | 0.991    | ± 0.000662 | 11      | 0.997    | ± 0.000156 |
| 2       | 0.998    | ± 8.9e-05  | 12      | 0.998    | ± 7.26e-05 |
| 3       | 0.998    | ± 7.97e-05 | 13      | 0.998    | ± 9.89e-05 |
| 4       | 0.991    | ± 0.000559 | 14      | 0.998    | ± 5.64e-05 |
| 5       | 0.997    | ± 0.000197 | 15      | 0.998    | ± 0.000118 |
| 6       | 0.994    | ± 0.00018  | 16      | 0.982    | ± 0.00376  |
| 7       | 0.976    | ± 0.00421  | 17      | 0.956    | ± 0.0121   |
| 8       | 0.998    | ± 6.71e-05 | 18      | 0.995    | ± 0.000286 |
| 9       | 0.998    | ± 8.32e-05 | 19      | 0.996    | ± 0.000122 |

| Qubit n | Fidelity | Error Bars | Qubit n | Fidelity | Error Bars |
|---------|----------|------------|---------|----------|------------|
| 0       | 0.997    | ± 0.000131 | 10      | 0.989    | ± 0.000582 |
| 1       | 0.998    | ± 7.47e-05 | 11      | 0.996    | ± 0.000218 |
| 2       | 0.998    | ± 7.57e-05 | 12      | 0.998    | ± 0.000126 |
| 3       | 0.995    | ± 0.000163 | 13      | 0.998    | ± 7.15e-05 |
| 4       | 0.996    | ± 0.000185 | 14      | 0.985    | ± 0.00183  |
| 5       | 0.995    | ± 0.000147 | 15      | 0.995    | ± 0.000469 |
| 6       | 0.998    | ± 8.34e-05 | 16      | 0.993    | ± 0.000558 |
| 7       | 0.887    | ± 0.0287   | 17      | 0.99     | ± 0.000502 |
| 8       | 0.999    | ± 4.48e-05 | 18      | 0.997    | ± 0.00026  |
| 9       | 0.998    | ± 7.2e-05  | 19      | 0.998    | ± 9.05e-05 |

## 8. Mermin

Mermin's algorithm for 3 qubits.

| Runtime    | Qubits   | Max   |
|------------|----------|-------|
| 227.53 sec | 10, 5, 6 | 3.358 |



| Runtime  | Qubits  | Max    |
|----------|---------|--------|
| 1.02 sec | 0, 2, 3 | -3.316 |



## 9. Grover - 2 qubits

Grover's algorithm for 2 qubits executed on sinq20 backend with 1000 shots per circuit. We measure the success rate of finding the target state '11' for each pair of qubits in  $[[5, 6]]$ .

| Runtime  | Qubits | Max prob |
|----------|--------|----------|
| 3.13 sec | 5, 6   | 0.874    |



| Runtime  | Qubits | Max prob |
|----------|--------|----------|
| 2.36 sec | 2, 3   | 0.89     |



## 10. Grover - 3 qubits

Grover's algorithm for 3 qubits executed on `siq20` backend with 1000 shots per circuit. We measure the success rate of finding the target state '111' for each pair of qubits in `[[10, 5], [5, 6]]`.

| Runtime  | Qubits   | Max prob |
|----------|----------|----------|
| 2.44 sec | 10, 6, 5 | 0.225    |



| Runtime  | Qubits  | Max prob |
|----------|---------|----------|
| 2.43 sec | 0, 2, 3 | 0.385    |



## 11. GHZ state preparation

GHZ circuit with 3 qubits executed on `siq20` backend with 1000 shots.

| Runtime  | Qubits   | Success |
|----------|----------|---------|
| 3.17 sec | 10, 5, 6 | 0.903   |



| Runtime  | Qubits  | Success |
|----------|---------|---------|
| 3.95 sec | 0, 3, 2 | 0.910   |

